Project 2 Analysis

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Data Import

```r
# Data import
# Using relative path assuming the working directory is set to the file's location
menu <- read.csv("menu.csv")
print(head(menu))
```

```
##    Category                             Item   Serving.Size Calories
## 1 Breakfast                     Egg McMuffin 4.8 oz (136 g)      300
## 2 Breakfast                Egg White Delight 4.8 oz (135 g)      250
## 3 Breakfast                 Sausage McMuffin 3.9 oz (111 g)      370
## 4 Breakfast        Sausage McMuffin with Egg 5.7 oz (161 g)      450
## 5 Breakfast Sausage McMuffin with Egg Whites 5.7 oz (161 g)      400
## 6 Breakfast             Steak & Egg McMuffin 6.5 oz (185 g)      430
##   Calories.from.Fat Total.Fat Total.Fat....Daily.Value. Saturated.Fat
## 1               120        13                        20             5
## 2                70         8                        12             3
## 3               200        23                        35             8
## 4               250        28                        43            10
## 5               210        23                        35             8
## 6               210        23                        36             9
##   Saturated.Fat....Daily.Value. Trans.Fat Cholesterol
## 1                            25         0         260
## 2                            15         0          25
## 3                            42         0          45
## 4                            52         0         285
## 5                            42         0          50
## 6                            46         1         300
##   Cholesterol....Daily.Value. Sodium Sodium....Daily.Value. Carbohydrates
## 1                          87    750                     31            31
## 2                           8    770                     32            30
## 3                          15    780                     33            29
## 4                          95    860                     36            30
## 5                          16    880                     37            30
## 6                         100    960                     40            31
##   Carbohydrates....Daily.Value. Dietary.Fiber Dietary.Fiber....Daily.Value.
## 1                            10             4                            17
## 2                            10             4                            17
## 3                            10             4                            17
## 4                            10             4                            17
```

```
## 5                            10             4                            17
## 6                            10             4                            18
##   Sugars Protein Vitamin.A....Daily.Value. Vitamin.C....Daily.Value.
## 1      3      17                       10                         0
## 2      3      18                        6                         0
## 3      2      14                        8                         0
## 4      2      21                       15                         0
## 5      2      21                        6                         0
## 6      3      26                       15                         2
##   Calcium....Daily.Value. Iron....Daily.Value.
## 1                      25                   15
## 2                      25                    8
## 3                      25                   10
## 4                      30                   15
## 5                      25                   10
## 6                      30                   20
```

```r
str(menu)
```

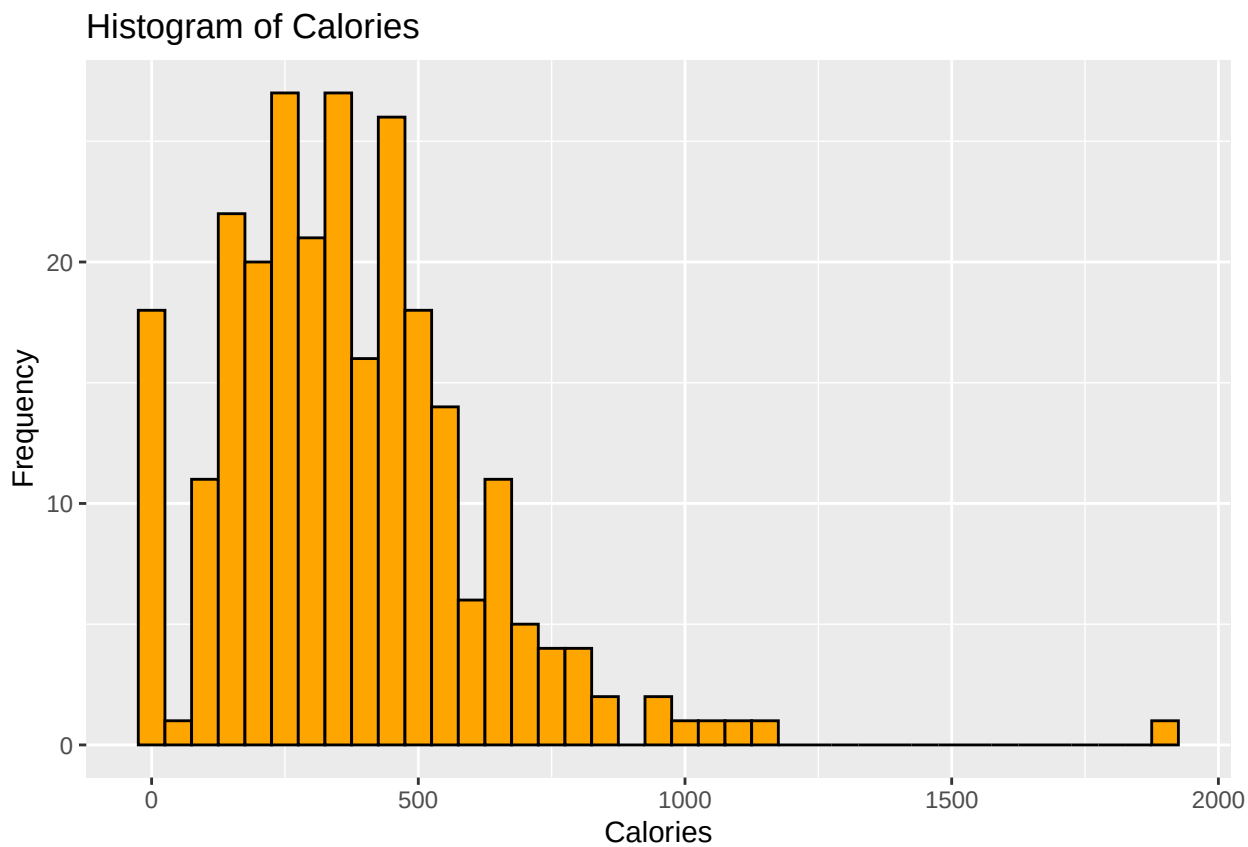
```
## 'data.frame':    260 obs. of  24 variables:
##  $ Category                     : chr  "Breakfast" "Breakfast" "Breakfast" "Breakfast" ...
##  $ Item                         : chr  "Egg McMuffin" "Egg White Delight" "Sausage McMuffin" "Sausag
##  $ Serving.Size                 : chr  "4.8 oz (136 g)" "4.8 oz (135 g)" "3.9 oz (111 g)" "5.7 oz (1
##  $ Calories                     : int  300 250 370 450 400 430 460 520 410 470 ...
##  $ Calories.from.Fat            : int  120 70 200 250 210 210 230 270 180 220 ...
##  $ Total.Fat                    : num  13 8 23 28 23 23 26 30 20 25 ...
##  $ Total.Fat....Daily.Value.    : int  20 12 35 43 35 36 40 47 32 38 ...
##  $ Saturated.Fat                : num  5 3 8 10 8 9 13 14 11 12 ...
##  $ Saturated.Fat....Daily.Value.: int  25 15 42 52 42 46 65 68 56 59 ...
##  $ Trans.Fat                    : num  0 0 0 0 0 1 0 0 0 0 ...
##  $ Cholesterol                  : int  260 25 45 285 50 300 250 250 35 35 ...
##  $ Cholesterol....Daily.Value.  : int  87 8 15 95 16 100 83 83 11 11 ...
##  $ Sodium                       : int  750 770 780 860 880 960 1300 1410 1300 1420 ...
##  $ Sodium....Daily.Value.       : int  31 32 33 36 37 40 54 59 54 59 ...
##  $ Carbohydrates                : int  31 30 29 30 30 31 38 43 36 42 ...
##  $ Carbohydrates....Daily.Value.: int  10 10 10 10 10 10 13 14 12 14 ...
##  $ Dietary.Fiber                : int  4 4 4 4 4 4 2 3 2 3 ...
##  $ Dietary.Fiber....Daily.Value.: int  17 17 17 17 17 18 7 12 7 12 ...
##  $ Sugars                       : int  3 3 2 2 2 3 3 4 3 4 ...
##  $ Protein                      : int  17 18 14 21 21 26 19 19 20 20 ...
##  $ Vitamin.A....Daily.Value.    : int  10 6 8 15 6 15 10 15 2 6 ...
##  $ Vitamin.C....Daily.Value.    : int  0 0 0 0 0 2 8 8 8 8 ...
##  $ Calcium....Daily.Value.      : int  25 25 25 30 25 30 15 20 15 15 ...
##  $ Iron....Daily.Value.         : int  15 8 10 15 10 20 15 20 10 15 ...
```

Histogram of Calories

```r
# Histogram of calories
plot1 <- ggplot(menu, aes(x = Calories)) +
  geom_histogram(binwidth = 50, fill = "orange", color = "black") +
  labs(
```

```
    title = "Histogram of Calories",
    x = "Calories",
    y = "Frequency"
  )

print(plot1)
```

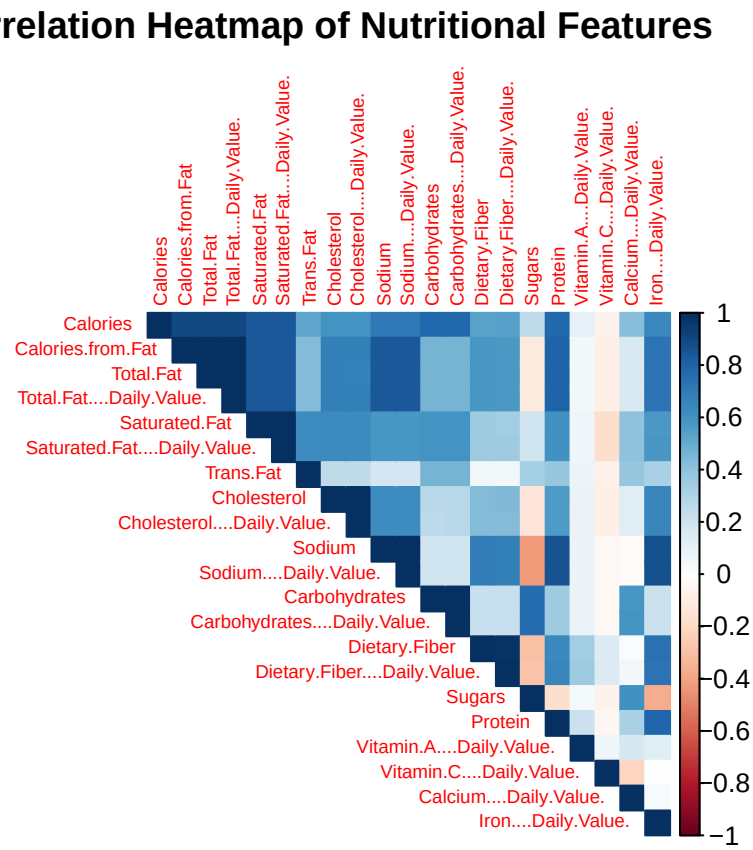


Correlation Analysis

```
# Selecting only numeric columns
numeric_menu <- menu %>% select(where(is.numeric))

# Computed correlation matrix
corr_matrix <- cor(numeric_menu, use = "complete.obs")

# Corr Heatmap
corrplot(corr_matrix,
  method = "color", type = "upper",
  tl.cex = 0.6, number.cex = 0.5,
  title = "Correlation Heatmap of Nutritional Features",
  mar = c(0, 0, 2, 0)
)
```



Top Correlations with Calories

```
# Correlations matrix with Calories
cal_corr <- corr_matrix[, "Calories"]

# Remove Calories itself
cal_corr <- cal_corr[names(cal_corr) != "Calories"]

# Descending sort
sorted_corr <- sort(cal_corr, decreasing = TRUE)

# Dropping features with repeated information
is_dv <- grepl("Daily.Value", names(sorted_corr), ignore.case = TRUE)
is_cfat <- names(sorted_corr) == "Calories.from.Fat"
clean_corr <- sorted_corr[!is_dv & !is_cfat]

# Top correlations
print(clean_corr[1:5])
```

```
##     Total.Fat Saturated.Fat       Protein Carbohydrates        Sodium
##     0.9044092     0.8455636     0.7878475     0.7815395     0.7123087
```

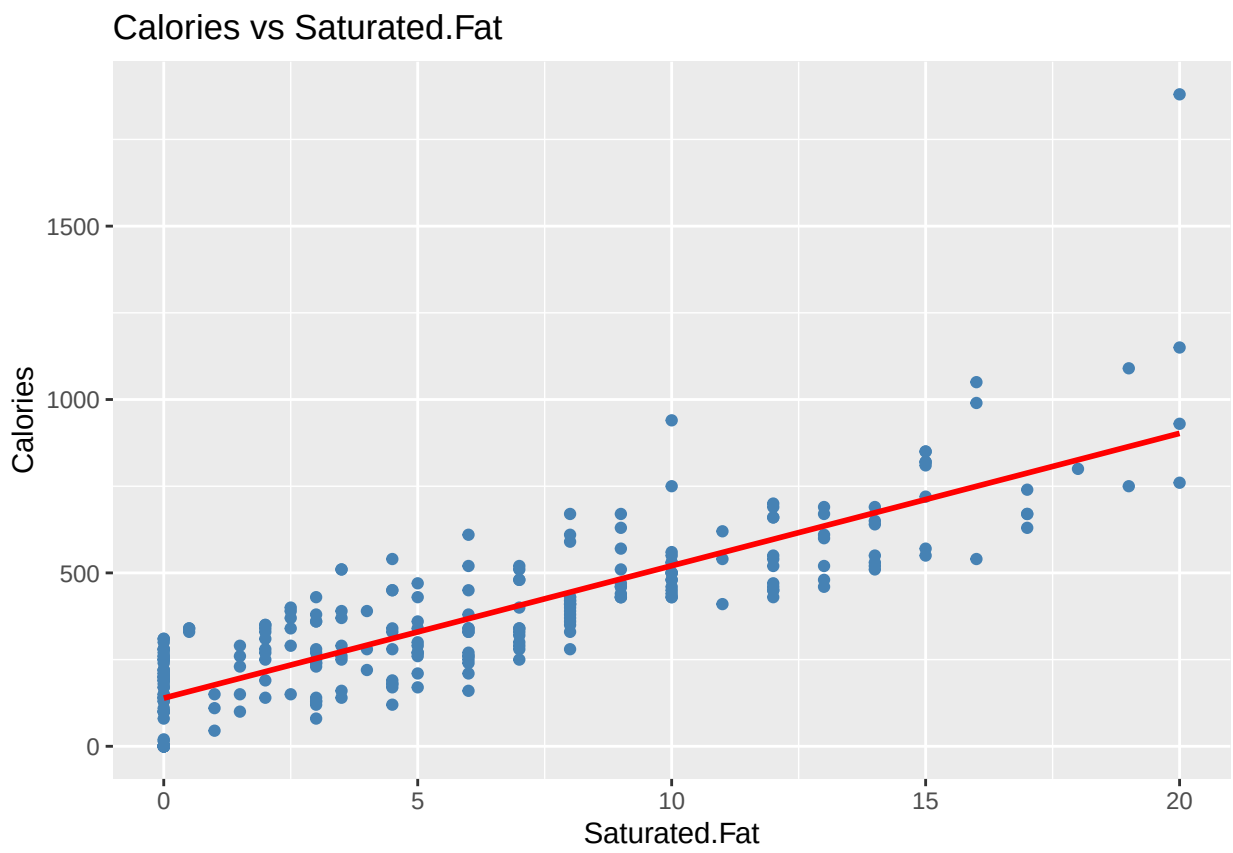
```r
# Negative correlations
neg_corr <- sorted_corr[sorted_corr < 0]
print(neg_corr)
```

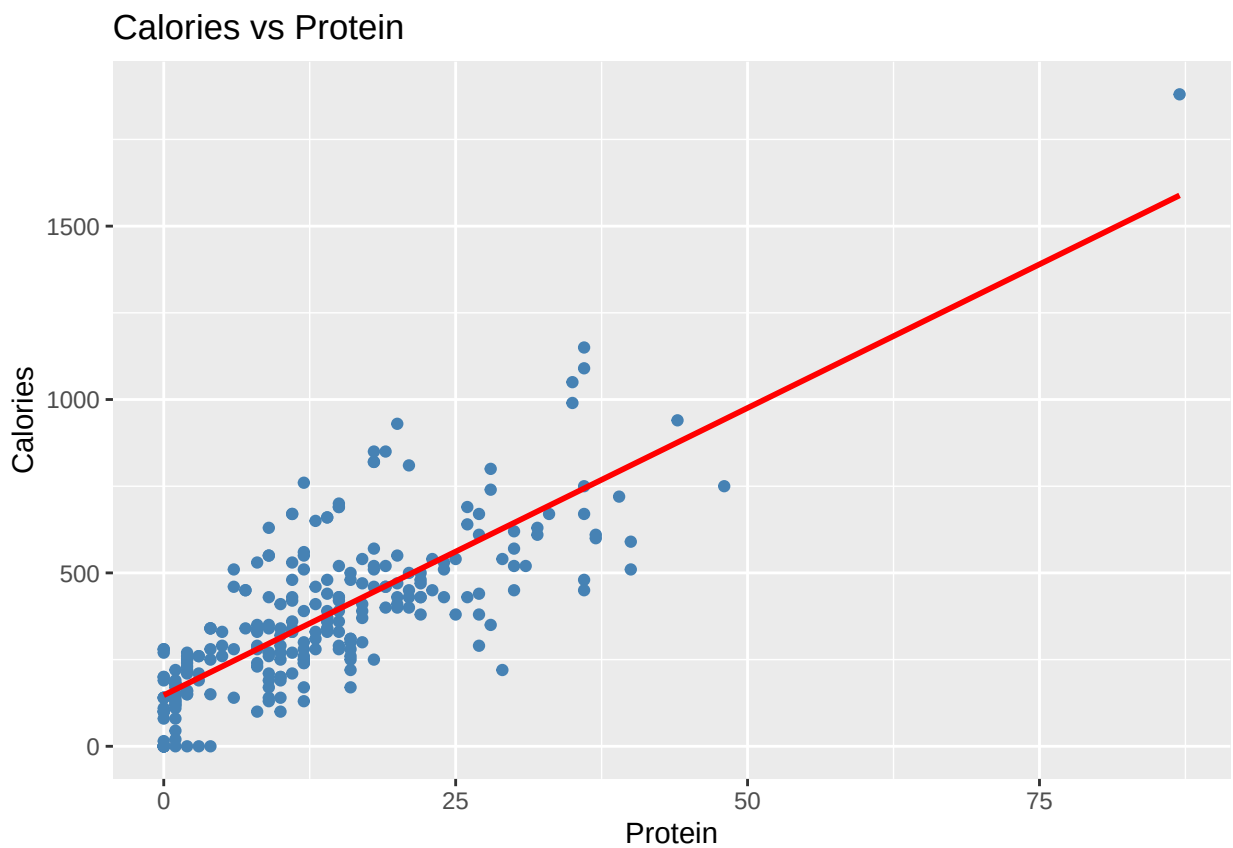
```
## Vitamin.C....Daily.Value.
##                -0.06874704
```

Feature Visualization

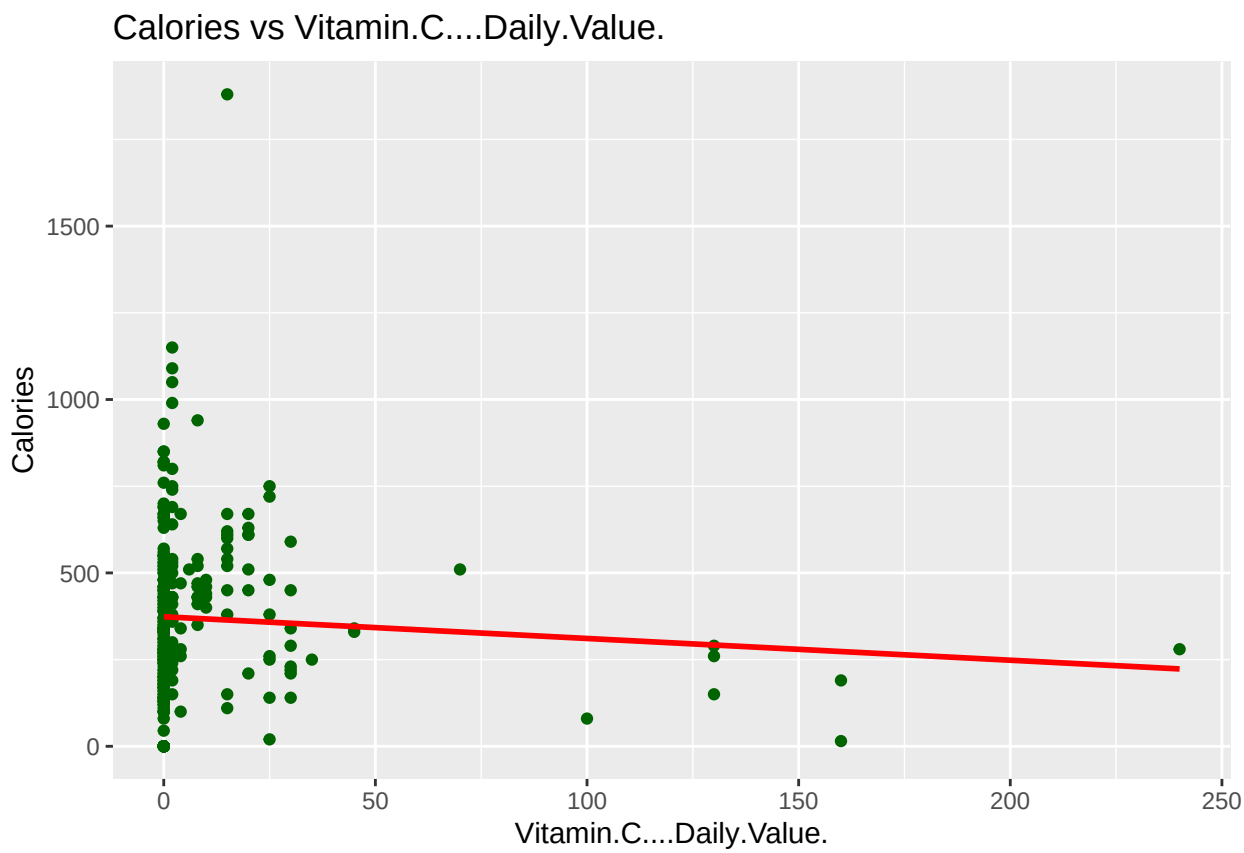
```r
# Feature selection
features_1c <- c("Saturated.Fat", "Protein")
features_1d <- c("Vitamin.C....Daily.Value.")

# Scatter plot for 1(c)
for (feat in features_1c) {
  print(
    ggplot(menu, aes_string(x = feat, y = "Calories")) +
      geom_point(color = "steelblue") +
      geom_smooth(method = "lm", se = FALSE, color = "red") +
      labs(
        title = paste("Calories vs", feat),
        x = feat,
        y = "Calories"
      )
  )
}
```

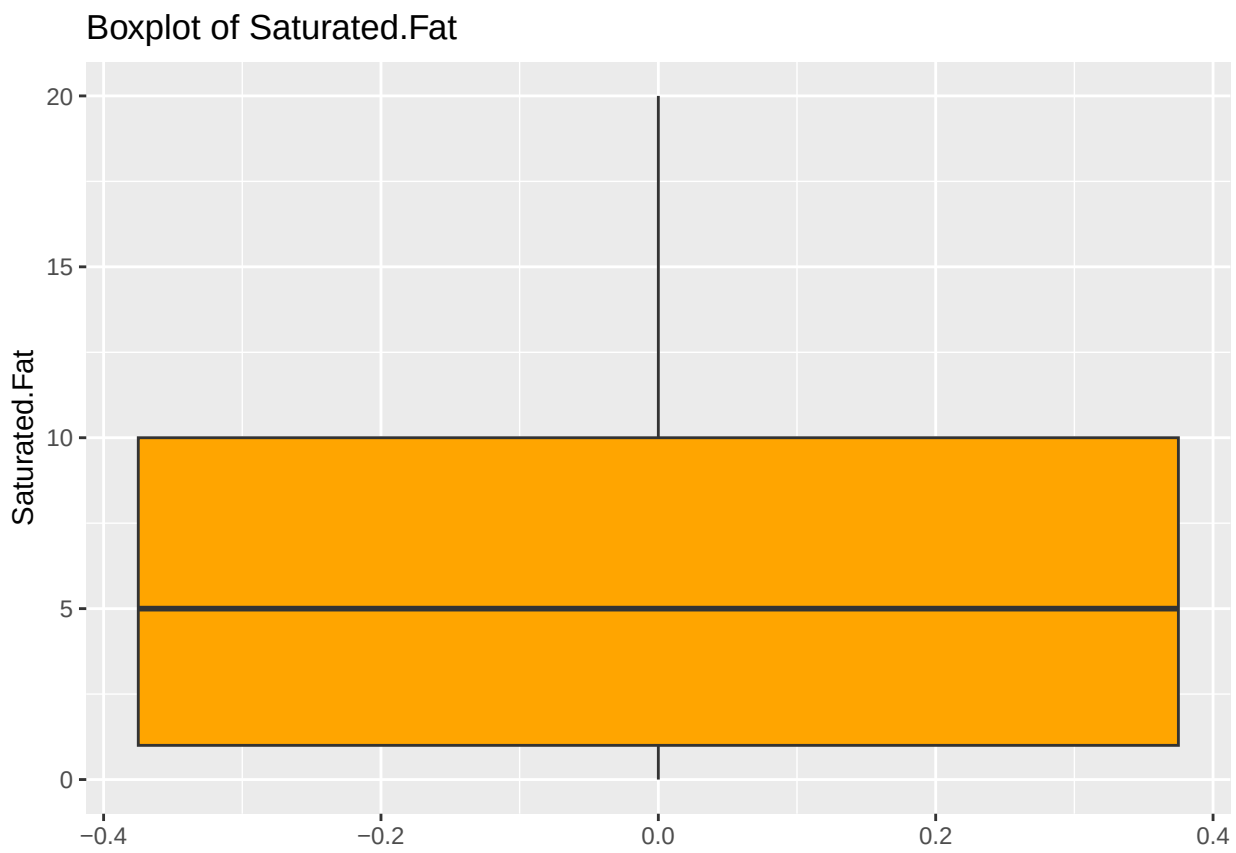

Calories vs Saturated.Fat
Calories
1500
1000
500
0
0
5
10
15
20
Saturated.Fat

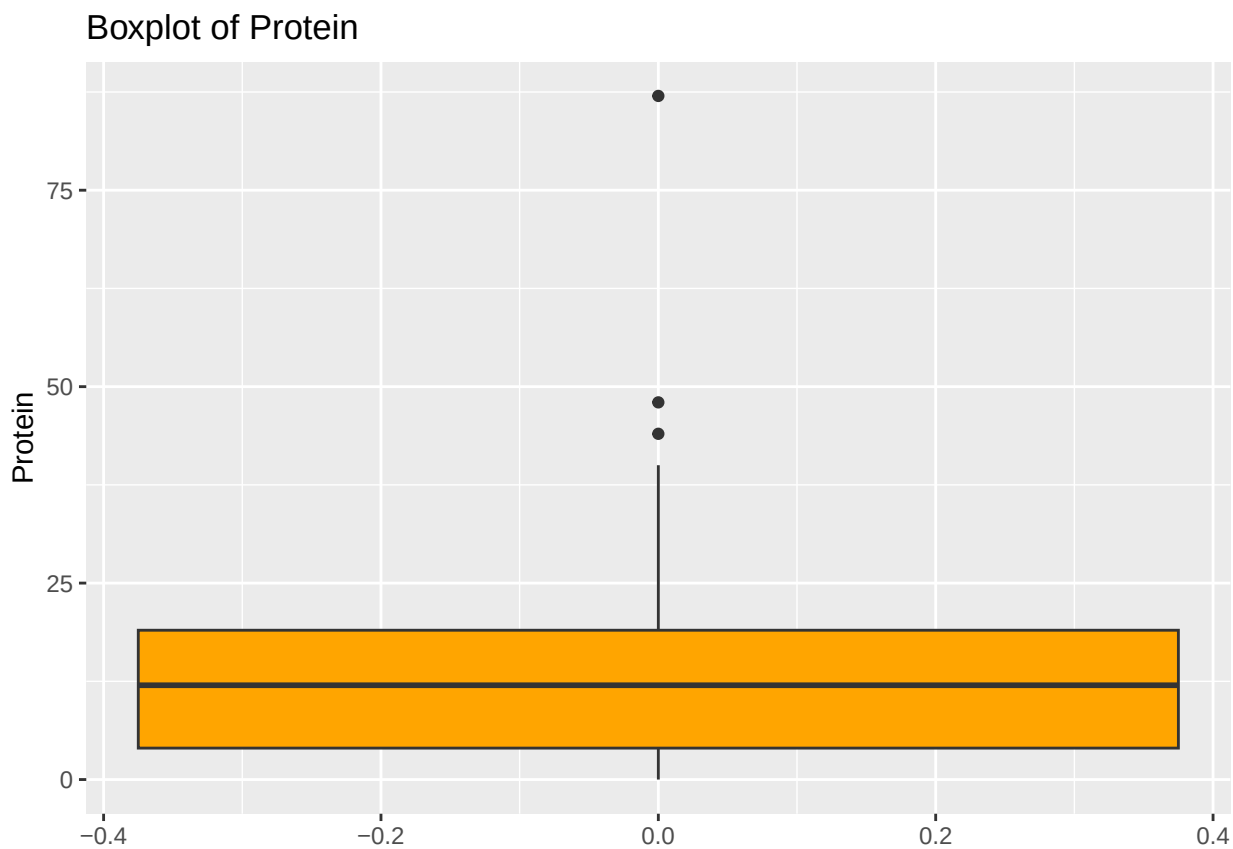



```r
# Scatter plot for 1(d)
for (feat in features_1d) {
  print(
    ggplot(menu, aes_string(x = feat, y = "Calories")) +
      geom_point(color = "darkgreen") +
      geom_smooth(method = "lm", se = FALSE, color = "red") +
      labs(
        title = paste("Calories vs", feat),
        x = feat,
        y = "Calories"
      )
  )
}
```

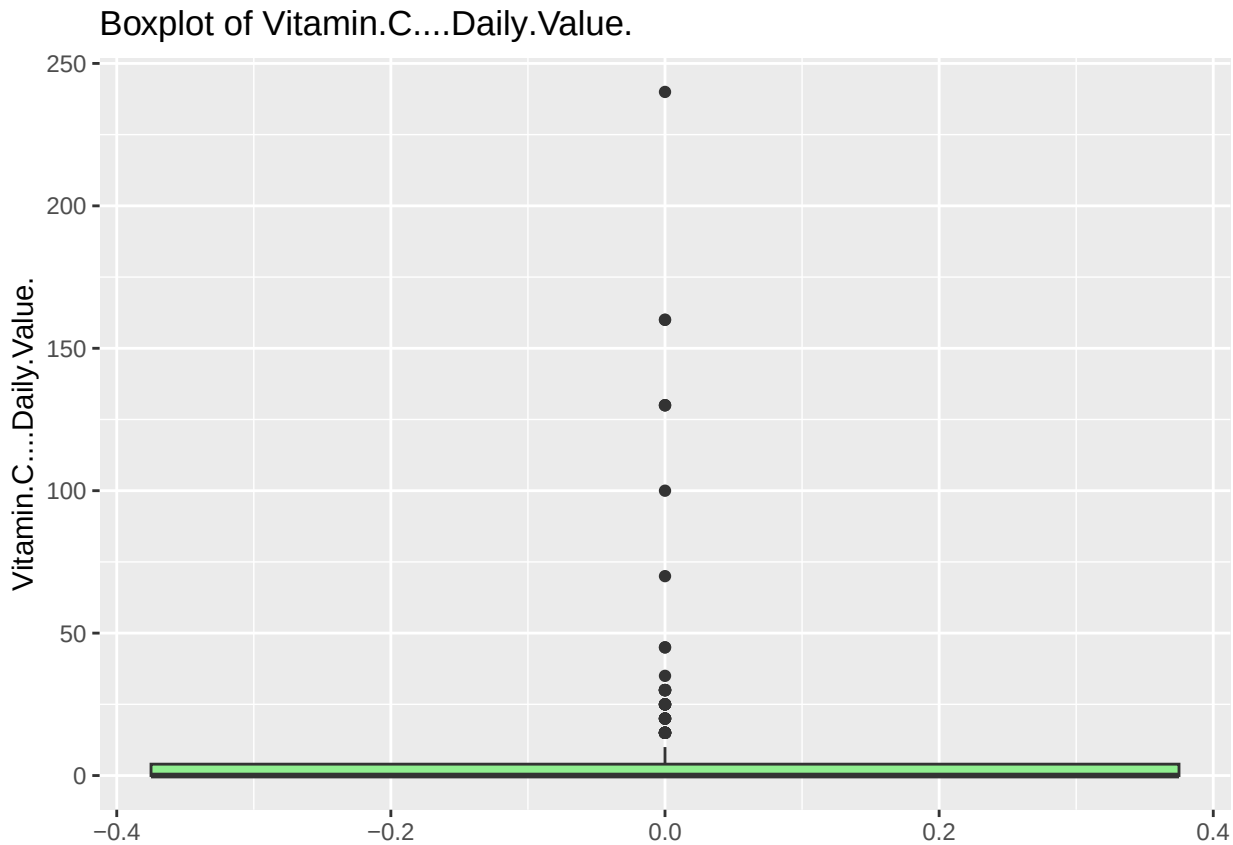



```r
# Boxplot for 1(c)
for (feat in features_1c) {
  print(
    ggplot(menu, aes_string(y = feat)) +
      geom_boxplot(fill = "orange") +
      labs(
        title = paste("Boxplot of", feat),
        y = feat
      )
  )
}
```


Boxplot of Saturated.Fat
Saturated.Fat
20
15
10
5
0
−0.4
−0.2
0.0
0.2
0.4




```r
# Boxplot for 1(d)
for (feat in features_1d) {
  print(
    ggplot(menu, aes_string(y = feat)) +
      geom_boxplot(fill = "lightgreen") +
      labs(
        title = paste("Boxplot of", feat),
        y = feat
      )
  )
}
```



Outlier Detection and Imputation

```
# Select numeric columns
numeric_menu <- menu[sapply(menu, is.numeric)]

# Compute median and sd
medians <- apply(numeric_menu, 2, median, na.rm = TRUE)
sds <- apply(numeric_menu, 2, sd, na.rm = TRUE)

print("Median values for all numerical features:")
```

```
## [1] "Median values for all numerical features:"
```

```
print(medians)
```

```
##                       Calories             Calories.from.Fat
##                          340.0                         100.0
##                      Total.Fat     Total.Fat....Daily.Value.
##                           11.0                          17.0
##                  Saturated.Fat Saturated.Fat....Daily.Value.
##                            5.0                          24.0
##                      Trans.Fat                   Cholesterol
##                            0.0                          35.0
```

```
##   Cholesterol....Daily.Value.                        Sodium
##                          11.0                         190.0
##        Sodium....Daily.Value.                 Carbohydrates
##                           8.0                          44.0
## Carbohydrates....Daily.Value.                 Dietary.Fiber
##                          15.0                           1.0
## Dietary.Fiber....Daily.Value.                        Sugars
##                           5.0                          17.5
##                       Protein     Vitamin.A....Daily.Value.
##                          12.0                           8.0
##     Vitamin.C....Daily.Value.       Calcium....Daily.Value.
##                           0.0                          20.0
##          Iron....Daily.Value.
##                           4.0
```

```r
print("Standard Deviations:")
```

```
## [1] "Standard Deviations:"
```

```r
print(sds)
```

```
##                      Calories             Calories.from.Fat
##                   240.2698865                   127.8759141
##                     Total.Fat     Total.Fat....Daily.Value.
##                    14.2059980                    21.8851988
##                 Saturated.Fat Saturated.Fat....Daily.Value.
##                     5.3218732                    26.6392088
##                     Trans.Fat                   Cholesterol
##                     0.4291329                    87.2692575
##   Cholesterol....Daily.Value.                        Sodium
##                    29.0916528                   577.0263230
##        Sodium....Daily.Value.                 Carbohydrates
##                    24.0349538                    28.2522317
## Carbohydrates....Daily.Value.                 Dietary.Fiber
##                     9.4195443                     1.5677171
## Dietary.Fiber....Daily.Value.                        Sugars
##                     6.3070573                    28.6797967
##                       Protein     Vitamin.A....Daily.Value.
##                    11.4261462                    24.3663807
##     Vitamin.C....Daily.Value.       Calcium....Daily.Value.
##                    26.3455423                    17.0199533
##          Iron....Daily.Value.
##                     8.7232633
```

```r
# Copy numeric data
clean_menu <- numeric_menu

# Function to replace outliers with NaN
replace_outliers <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1
```

```r
  lower <- Q1 - 1.5 * IQR_val
  upper <- Q3 + 1.5 * IQR_val

  outliers <- (x < lower | x > upper)
  x[outliers] <- NaN

  return(list(values = x, count = sum(outliers)))
}

# Apply to all numeric features
outlier_results <- lapply(clean_menu, replace_outliers)

# Extract cleaned data
clean_menu <- as.data.frame(lapply(outlier_results, function(x) x$values))

# Count NaNs introduced per feature
nan_counts <- sapply(outlier_results, function(x) x$count)
print("NaN counts introduced per feature:")
```

```
## [1] "NaN counts introduced per feature:"
```

```r
print(nan_counts)
```

```
##                      Calories             Calories.from.Fat
##                             6                             4
##                     Total.Fat     Total.Fat....Daily.Value.
##                             4                             4
##                 Saturated.Fat Saturated.Fat....Daily.Value.
##                             0                             0
##                     Trans.Fat                   Cholesterol
##                            56                            18
##   Cholesterol....Daily.Value.                        Sodium
##                            18                             5
##        Sodium....Daily.Value.                 Carbohydrates
##                             5                            17
## Carbohydrates....Daily.Value.                 Dietary.Fiber
##                            16                             0
## Dietary.Fiber....Daily.Value.                        Sugars
##                             4                             4
##                       Protein     Vitamin.A....Daily.Value.
##                             3                            17
##     Vitamin.C....Daily.Value.       Calcium....Daily.Value.
##                            46                             2
##          Iron....Daily.Value.
##                             2
```

```r
# Replace NaN with mean
filled_menu <- clean_menu

for (col in names(filled_menu)) {
  mean_val <- mean(filled_menu[[col]], na.rm = TRUE)
  filled_menu[[col]][is.nan(filled_menu[[col]])] <- mean_val
```

```r
}

# Recomputing median and sd
medians_after <- apply(filled_menu, 2, median)
sds_after <- apply(filled_menu, 2, sd)

print("Median and SD after imputation:")
```

```
## [1] "Median and SD after imputation:"
```

```r
print("Median:")
```

```
## [1] "Median:"
```

```r
print(medians_after)
```

```
##                      Calories             Calories.from.Fat
##                     340.00000                     100.00000
##                     Total.Fat     Total.Fat....Daily.Value.
##                      11.00000                      17.00000
##                 Saturated.Fat Saturated.Fat....Daily.Value.
##                      5.00000                      24.00000
##                     Trans.Fat                   Cholesterol
##                      0.00000                      34.66942
##   Cholesterol....Daily.Value.                       Sodium
##                     11.00000                     190.00000
##        Sodium....Daily.Value.                 Carbohydrates
##                      8.00000                      42.27572
## Carbohydrates....Daily.Value.                 Dietary.Fiber
##                     14.18033                       1.00000
## Dietary.Fiber....Daily.Value.                        Sugars
##                      5.00000                      17.50000
##                      Protein     Vitamin.A....Daily.Value.
##                     12.00000                       8.00000
##     Vitamin.C....Daily.Value.       Calcium....Daily.Value.
##                      0.00000                      20.00000
##          Iron....Daily.Value.
##                      4.00000
```

```r
print("Standard Deviations:")
```

```
## [1] "Standard Deviations:"
```

```r
print(sds_after)
```

```
##                      Calories             Calories.from.Fat
##                    199.054754                    105.225212
##                     Total.Fat     Total.Fat....Daily.Value.
##                    11.692321                      17.983982
##                 Saturated.Fat Saturated.Fat....Daily.Value.
```

```
##                      5.321873                     26.639209
##                     Trans.Fat                   Cholesterol
##                      0.000000                     29.849918
##   Cholesterol....Daily.Value.                        Sodium
##                      9.913967                    498.403136
##        Sodium....Daily.Value.                 Carbohydrates
##                     20.754357                     20.484341
## Carbohydrates....Daily.Value.                 Dietary.Fiber
##                      6.964905                      1.567717
## Dietary.Fiber....Daily.Value.                        Sugars
##                      5.705188                     26.252703
##                       Protein     Vitamin.A....Daily.Value.
##                     10.052272                      7.296993
##     Vitamin.C....Daily.Value.       Calcium....Daily.Value.
##                      2.077643                     16.461288
##          Iron....Daily.Value.
##                      8.245846
```